EVALUASI IOT-MQTT

DISUSUN OLEH : KELOMPOK 9

Anggota Kelompok

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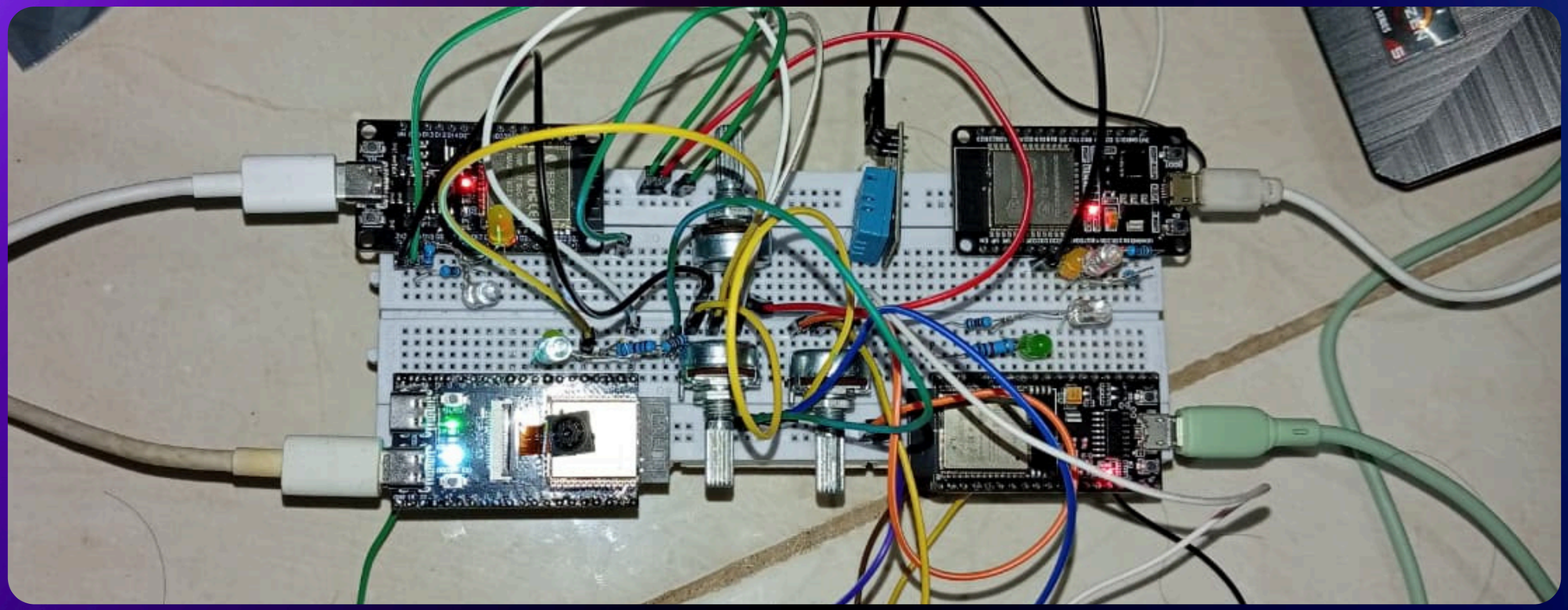
Latar Belakang

Fokus Utama: Implementasi arsitektur IoT berbasis protokol MQTT untuk monitoring dan kendali terpusat di 4 ruangan.

Perangkat Sistem: Menggunakan mikrokontroler ESP32 sebagai node yang membaca data sensor (DHT & Potensiometer) dan mengontrol aktuator (Lampu LED).
Komunikasi Data: Menggunakan topik spesifik (contoh: pens/klp9/...) dilengkapi parameter QoS dan Retain untuk menjamin keandalan transmisi pesan.

Fokus Evaluasi: Pengujian antarmuka dashboard melalui Node-RED, verifikasi aliran data via aplikasi MQTTX, dan penyelesaian kendala konektivitas IP broker.

Rangkaian



Pin yang Digunakan

Tabel Pin yang Dipakai Ruangan 1, 2, 3, dan 4

Ruangan	Kategori	Sensor / Input	LED Lampu 1	LED Lampu 2
Ruang 1	Komponen	Potensiometer 10 kΩ	LED lampu 1 + resistor 220 Ω	LED lampu 2 + resistor 220 Ω
	Pin ESP32	GPIO32	GPIO16	GPIO19
Ruang 2	Komponen	DHT	LED lampu 1 + resistor 220 Ω	LED lampu 2 + resistor 220 Ω
	Pin ESP32	GPIO32	GPIO25	GPIO33
Ruang 3	Komponen	Potensiometer 10 kΩ	LED lampu 1 + resistor 220 Ω	LED lampu 2 + resistor 220 Ω
	Pin ESP32	GPIO32	GPIO16	GPIO18
Ruang 4	Komponen	Potensiometer 10 kΩ	LED lampu 1 + resistor 220 Ω	LED lampu 2 + resistor 220 Ω
	Pin ESP32	GPIO6	GPIO16	GPIO18

Tabel topic final

Tabel Topic Final

Topic	Arah	Payload	QoS	Retain
pens/klp9/ruangN/temperature	ESP32 → Node-RED	°C, 1 Desimal, mis. 28.5	0	Tidak
pens/klp9/ruangN/humidity	ESP32 → Node-RED	% 1 desimal, mis. 65.0	0	Tidak
pens/klp9/ruangN/lamp1	Node-RED → ESP32	1 = nyala 0 = mati	1	Tidak
pens/klp9/ruangN/lamp2	Node-RED → ESP32	1 = nyala 0 = mati	1	Tidak
pens/klp9/ruangN/status	LWT	Online atau offline	0	Ya

Kode Program

```
#include <WiFi.h>
#include <PubSubClient.h>
#include <DHT.h>
// Library: "DHT sensor library" by Adafruit (+ "Adafruit Unified Sensor")

// ===== KONFIGURASI JARINGAN =====
const char* WIFI_SSID = "yagimana";
const char* WIFI_PASS = "yabegitulah";
const char* BROKER_IP = "broker.emqx.io";

// ===== IDENTIFIKASI RUANGAN =====
const char* ROOM_NAME = "ruang2";

// ===== PEMETAAN PIN HARDWARE (ESP32 DevKit V1) =====
const int DHT_PIN = 32;  // Pin DATA DHT11
const int L1_PIN  = 16;  // LED Lampu 1
const int L2_PIN  = 19;  // LED Lampu 2

#define DHT_TYPE DHT11
DHT dht(DHT_PIN, DHT_TYPE);

WiFiClient net;
PubSubClient mqtt(net);
String clientId;

// Topic Buffers (pola pens/klp9/<room>/...)
char tTemp[50], tHum[50], tL1[50], tL2[50], tStatus[50];

unsigned long lastPublish = 0;
unsigned long lastWiFiRetry = 0;

// ===== CALLBACK HANDLING (SUBSCRIBE) =====
void callback(char* topic, byte* payload, unsigned int length) {
  String message = "";
  for (unsigned int i = 0; i < length; i++) {
    message += (char)payload[i];
  }
  message.trim();

  Serial.print("Pesan masuk [Topic: ");
  Serial.print(topic);
  Serial.print("] Value: ");
  Serial.println(message);

  int state = (message == "1") ? HIGH : LOW;

  if (strcmp(topic, tL1) == 0) {
    if (message == "1" || message == "0") {
      digitalWrite(L1_PIN, state);
      Serial.println(state == HIGH ? "Lampu 1 MENYALA" : "Lampu 1 MATI");
    }
  }
```

```
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  }
  message.trim();

  Serial.print("Pesan masuk [Topic: ");
  Serial.print(topic);
  Serial.print("] Value: ");
  Serial.println(message);

  int state = (message == "1") ? HIGH : LOW;

  if (strcmp(topic, tL1) == 0) {
    if (message == "1" || message == "0") {
      digitalWrite(L1_PIN, state);
      Serial.println(state == HIGH ? "Lampu 1 MENYALA" : "Lampu 1 MATI");
    }
  }
  else if (strcmp(topic, tL2) == 0) {
    if (message == "1" || message == "0") {
      digitalWrite(L2_PIN, state);
      Serial.println(state == HIGH ? "Lampu 2 MENYALA" : "Lampu 2 MATI");
    }
  }
}

void setup() {
  Serial.begin(9600);
  delay(1000);

  pinMode(L1_PIN, OUTPUT);
  pinMode(L2_PIN, OUTPUT);
  digitalWrite(L1_PIN, LOW);
  digitalWrite(L2_PIN, LOW);

  dht.begin();
```

```
  dht.begin();

  snprintf(tTemp,   sizeof(tTemp),   "pens/klp9/%s/temperature",
ROOM_NAME);
  snprintf(tHum,    sizeof(tHum),    "pens/klp9/%s/humidity", ROOM_NAME);
  snprintf(tL1,     sizeof(tL1),     "pens/klp9/%s/lamp1",    ROOM_NAME);
  snprintf(tL2,     sizeof(tL2),     "pens/klp9/%s/lamp2",    ROOM_NAME);
  snprintf(tStatus, sizeof(tStatus), "pens/klp9/%s/status",   ROOM_NAME);

  Serial.println("\nData Ruangan");
  Serial.println(tTemp);
  Serial.println(tHum);
  Serial.println(tL1);
  Serial.println(tL2);
  Serial.println(tStatus);
  Serial.println("_____");

  WiFi.mode(WIFI_STA);
  Serial.print("Connecting to WiFi: ");
  Serial.println(WIFI_SSID);
  WiFi.begin(WIFI_SSID, WIFI_PASS);

  clientId = "esp32-devkit-pens-" + WiFi.macAddress();

  mqtt.setServer(BROKER_IP, 1883);
  mqtt.setCallback(callback);
  mqtt.setKeepAlive(15);
}

void connectMQTT() {
  if (WiFi.status() != WL_CONNECTED) return;

  static unsigned long lastConnectAttempt = 0;
  if (millis() - lastConnectAttempt >= 5000) {
    lastConnectAttempt = millis();

    Serial.print("Attempting MQTT connection...");
    if (mqtt.connect(clientId.c_str(), tStatus, 0, true, "offline")) {
      Serial.println("connected!");
      mqtt.subscribe(tL1, 1);
      mqtt.subscribe(tL2, 1);
      mqtt.publish(tStatus, "online", true);
    } else {
      Serial.print("failed, rc=");
      Serial.print(mqtt.state());
      Serial.println(" try again in 5 seconds");
    }
  }
}

void loop() {
  // Penanganan koneksi WiFi
```

```
}

void loop() {
  // Penanganan koneksi WiFi
  if (WiFi.status() != WL_CONNECTED) {
    if (millis() - lastWiFiRetry >= 10000) {
      lastWiFiRetry = millis();
      Serial.println("WiFi disconnected. Reconnecting...");
      WiFi.disconnect();
      WiFi.begin(WIFI_SSID, WIFI_PASS);
    }
    return;
  }

  // Penanganan koneksi MQTT
  if (!mqtt.connected()) {
    connectMQTT();
    return;
  }

  mqtt.loop();

  // Kirim data sensor setiap 5 detik
  if (millis() - lastPublish >= 5000) {
    lastPublish = millis();

    // Baca DHT11
    float suhu = dht.readTemperature();  // Celsius
    float kelembapan = dht.readHumidity();

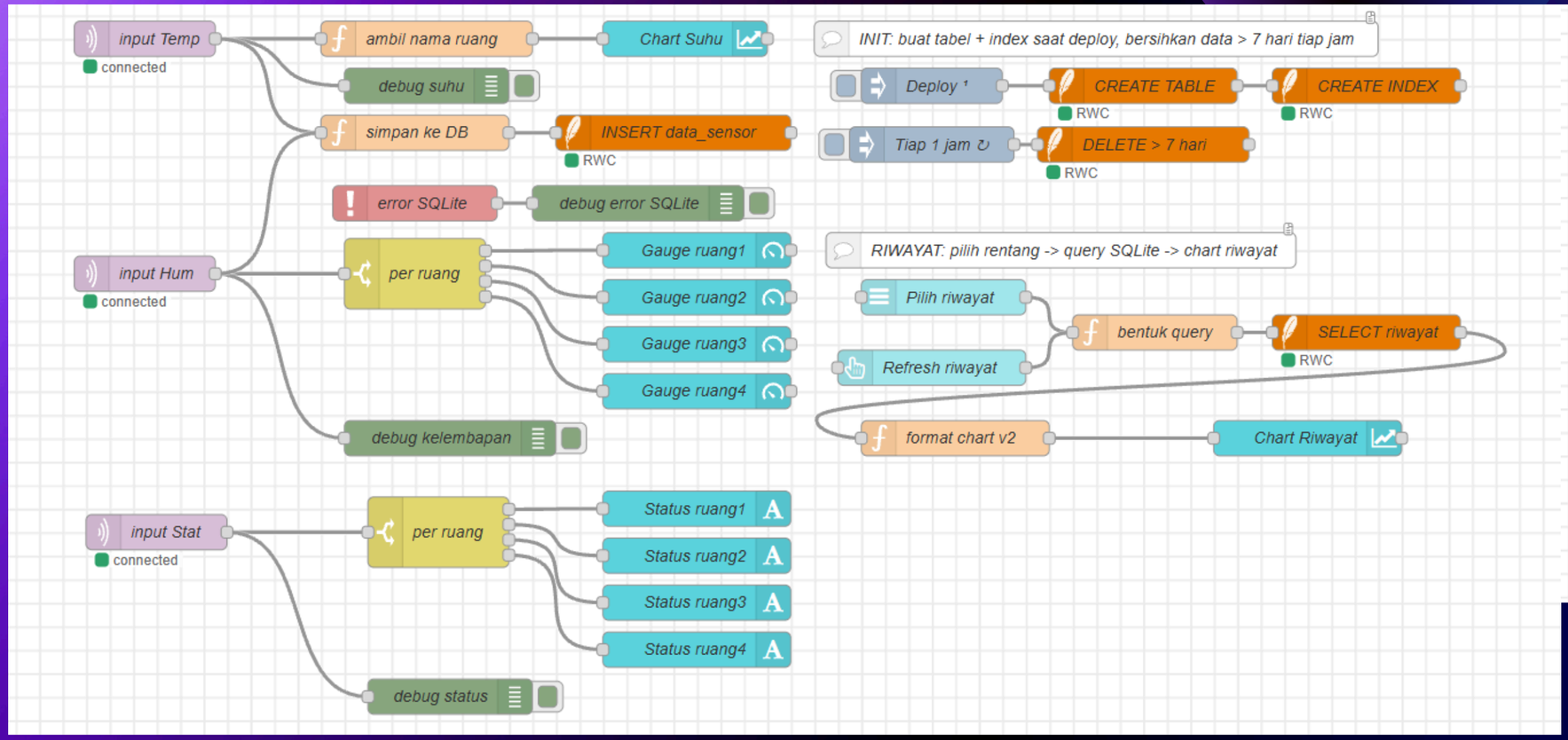
    // DHT11 bisa gagal dibaca (NaN) -> lewati publish agar tidak kirim data sampah
    if (isnan(suhu) || isnan(kelembapan)) {
      Serial.println("[ERROR] Gagal membaca sensor DHT11, cek wiring.");
      return;
    }

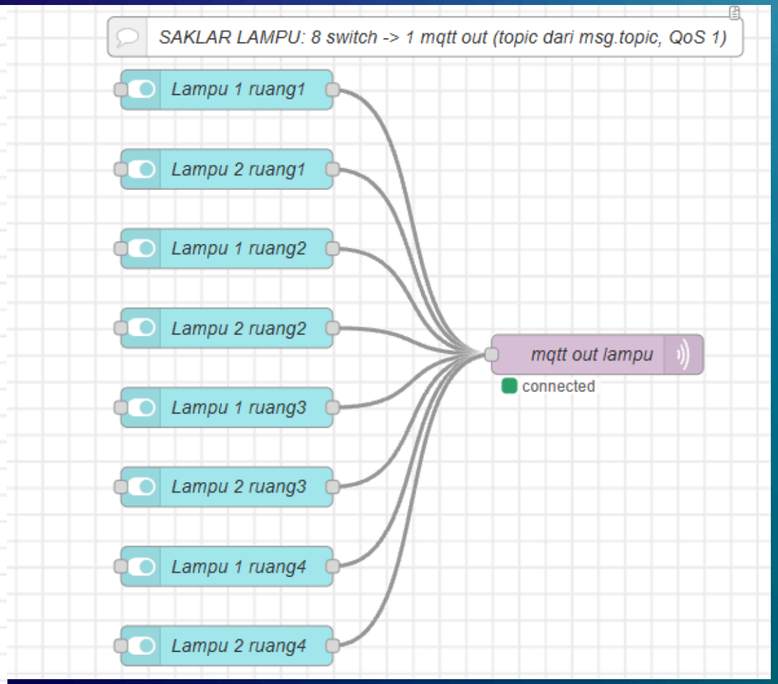
    char tempBuf[8];
    snprintf(tempBuf, sizeof(tempBuf), "%.1f", suhu);
    mqtt.publish(tTemp, tempBuf);

    char humBuf[8];
    snprintf(humBuf, sizeof(humBuf), "%.1f", kelembapan);
    mqtt.publish(tHum, humBuf);

    Serial.print("[Publish] Suhu (DHT11): ");
    Serial.print(tempBuf);
    Serial.print(" °C | Kelembapan (DHT11): ");
    Serial.print(humBuf);
    Serial.println(" %");
  }
}
```

Flow Node-Red





TERIMA KASIH